

NOT GATE

Combinational Analysis

File Edit Project Simulate Window Help

Inputs Outputs Table Expression Minimized

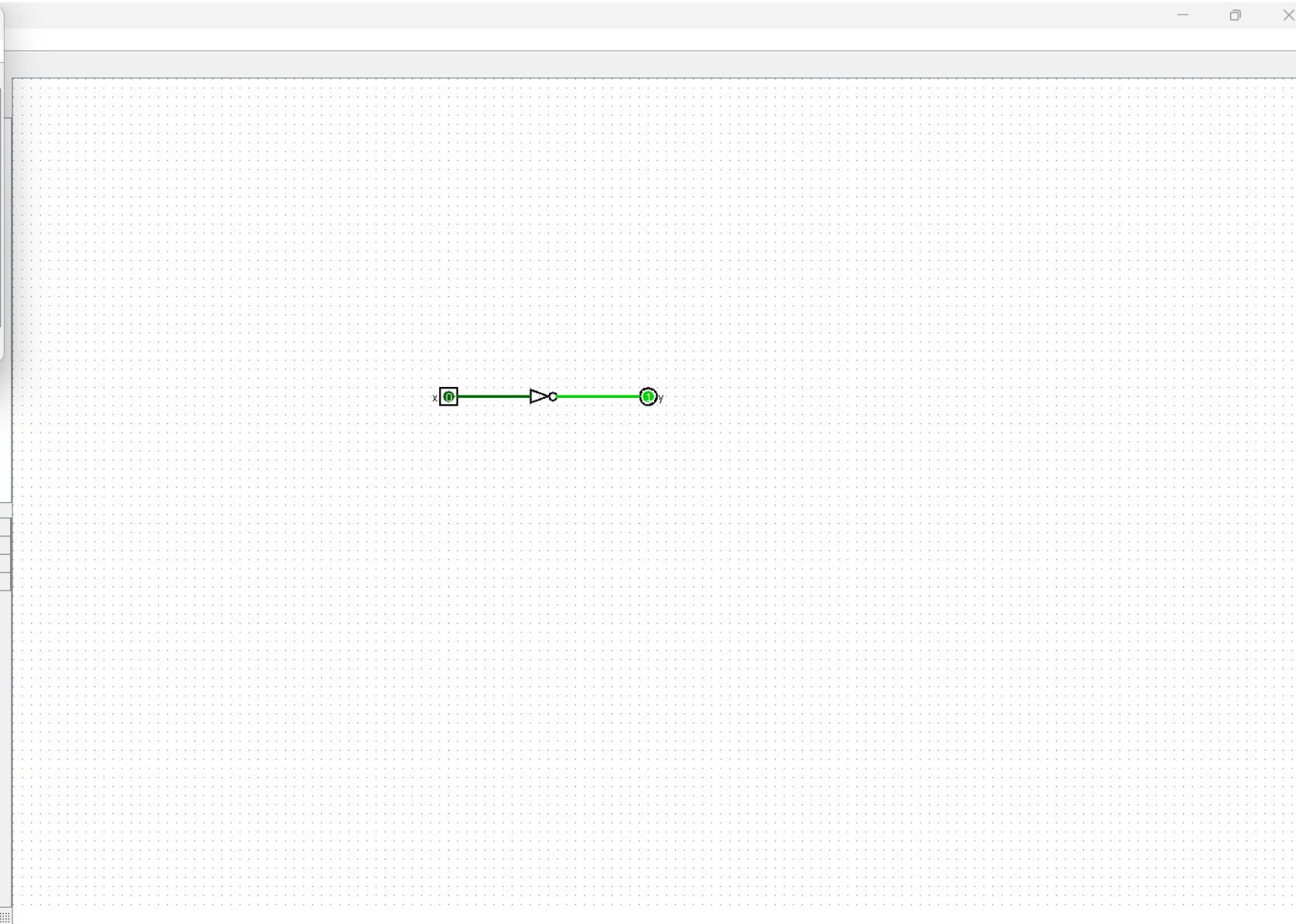
x	y
0	1
1	0

Build Circuit

Circuit: main

Circuit Name	main
Shared Label	
Shared Label Facing	East
Shared Label Font	SansSerif Plain 12

100%



AND GATE

Combinational Analysis

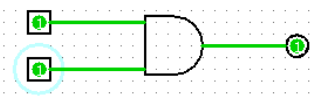
File Edit Project Simulate Window Help

Inputs Outputs Table Expression Minimized

a	b	x
0	0	0
0	1	0
1	0	0
1	1	1

Build Circuit

Pin	
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	
Label Location	West
Label Font	SansSerif Plain 12



OR GATE

Combinational Analysis

File Edit Project Simulate Window Help

Inputs Outputs **Table** Expression Minimized

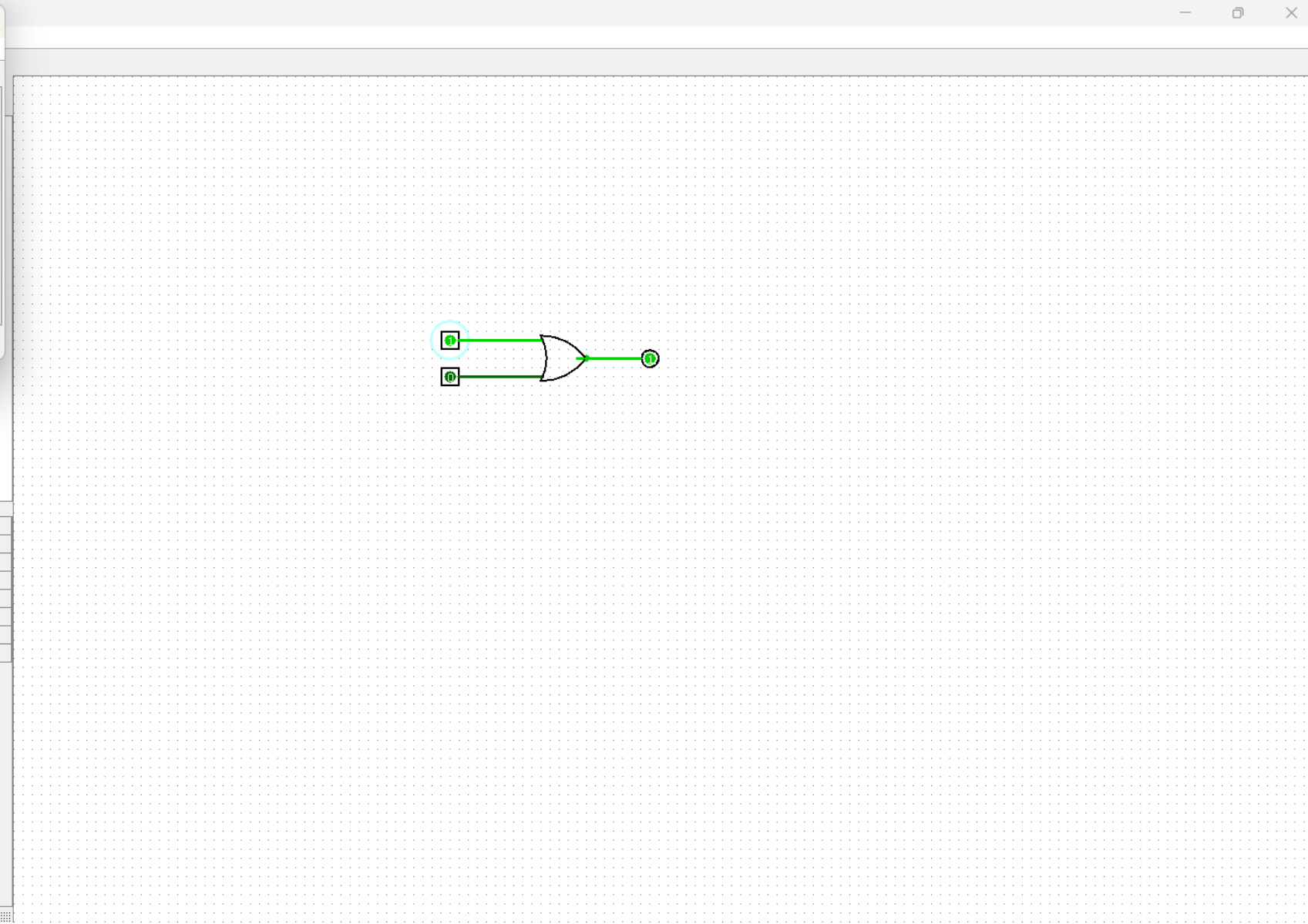
View and manipulate outputs' expressions.

a	b	x
0	0	0
0	1	1
1	0	1
1	1	1

Build Circuit

- Arithmetic
- Memory
- Input/Output
- Base

	Pin
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	
Label Location	West
Label Font	SansSerif Plain 12



100%

NAND GATE

Combinational Analysis

File Edit Project Simulate Window Help

Inputs Outputs **Table** Expression Minimized

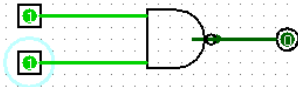
a	b	x
0	0	1
0	1	1
1	0	1
1	1	0

Build Circuit

- Arithmetic
- Memory
- Input/Output
- Base

Pin	
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	
Label Location	West
Label Font	SansSerif Plain 12

100%



NOR GATE

Combinational Analysis

File Edit Project Simulate Window Help

Inputs Outputs **Table** Expression Minimized

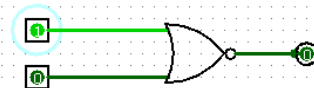
a	b	x
0	0	1
0	1	0
1	0	0
1	1	0

Build Circuit

- Arithmetic
- Memory
- Input/Output
- Base

Pin	
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	
Label Location	West
Label Font	SansSerif Plain 12

100%



XNOR GATE

Combinational Analysis

File Edit Project Simulate Window Help

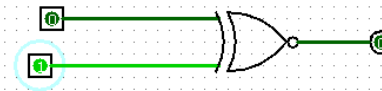
Inputs Outputs **Table** Expression Minimized

a	b	x
0	0	1
0	1	0
1	0	0
1	1	1

Build Circuit

- Arithmetic
- Memory
- Input/Output
- Base

	Pin
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	
Label Location	West
Label Font	SansSerif Plain 12



HALF ADDER

Combinational Analysis

File Edit Project Simulate Window Help

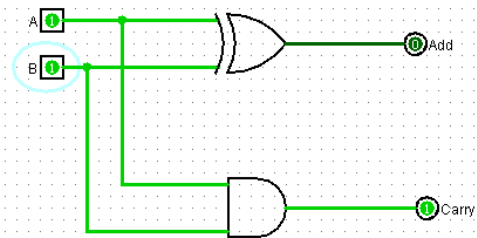
Inputs Outputs **Table** Expression Minimized

A	B	Add	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Build Circuit

Arithmetic
Memory
Input/Output
Base

Pin	
Facing	East
Output?	No
Data Bits	1
Three-state?	No
Pull Behavior	Unchanged
Label	B
Label Location	West
Label Font	SansSerif Plain 12



FULL ADDER

Combinational Analysis

File Edit Project Simulate Window Help

Inputs Outputs Table Expression Minimized

A	B	CIn	SUM	COut
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Build Circuit

- Arithmetic
- Memory
- Input/Output
- Base

Selection: Pin	
Facing	West
Output?	Yes
Data Bits	1
Three-state?	Yes
Pull Behavior	Unchanged
Label	C out
Label Location	East
Label Font	SansSerif Plain 12

